A Cylindrical Shell Discrete Element

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The development of the stiffness and consistent mass matrices for a finite cylindrical shell element is reported. The derivation of these matrices is based upon linear behavior and thin-shell assumptions. Expressing the assumed displacement state over the middle surface of the cylindrical shell element as products of one-dimensional, first-order Hermite interpolation formulas, it is possible to insure that the displacement state for the assembled set of cylindrical elements is geometrically admissible. Monotonic convergence of the total potential energy is therefore assured as the modeling is successively refined, since the assumed displacement pattern satisfies the admissibility requirements in the statement of the principle of minimum potential energy. A numerical example is included to demonstrate the effectiveness of this cylindrical shell discrete element.

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Nomenclature

D	= flexural rigidity
E	= modulus of elasticity
$G_{ij}^{(k)}$	= constants characterizing stiffness and mass matrices
h	= thickness of shell
$H_{kl}^{(1)}$	= first-order Hermite polynomials
K	= stiffness matrix
$K^{(u)}$, $K^{(v)}$, $K^{(w)}$, $K^{(vu)}$, $K^{(wu)}$, $K^{(wv)}$	= submatrices of the stiffness matrix
l	= length of cylindrical shell element
M	= consistent mass matrix
$M^{(u)}$, $M^{(v)}$, $M^{(w)}$	= submatrices of the mass matrix
N	= number of elements in an assembly
p_u, p_v, p_w	= middle surface tractions in directions of the displacements u, v, w, respectively
r	= radius of cylindrical shell element
s, θ, z	= coordinate directions
u, v, w	= middle surface displacements in the s, θ, z directions, respectively
u_{ij}, u_{si_j}, $u_{\theta ij}$, $u_{s\theta ij}$, v_{ij}, . . . , w_{ij}, . . .	= element nodal parameters
U	= element strain energy
W	= external work
X	= vector of nodal parameters
ϵ_s, ϵ_θ, $\epsilon_{s\theta}$	= strains a distance z from the middle surface
$\Delta\theta$	= subtended angle of cylindrical element
ν	= Poisson's ratio
Π_p	= total potential energy, $U - W$
$\Pi_p^{(k)}$	= potential energy of kth element

In recent years, the finite-element approach to the derivation of stiffness matrices for the analysis of plate and shell structures by the displacement method has received a vast amount of attention. As a result of the activity in this area of structural mechanics, the displacement method has experienced a more or less distinct evolution from the early 1950's to the current state-of-the-art. Initial efforts at finite element modeling viewed an aircraft structure as an assemblage of one-dimensional beams and torque boxes as introduced by Levy.[1] This capability was refined by Turner, Clough, Martin, and Topp[2] to include rectangular and triangular elements capable of representing two-dimensional in-plane behavior. Finite-element representations for rectangular plates in bending were developed by Pappenfuss[3] and Melosh.[4]

Until recently, additional research on the direct stiffness method has been directed toward refining rectangular and triangular plate elements in an attempt to rectify certain discrepancies that seemed to crop up incessantly, namely, application of the various stiffness matrices yielded results that exhibited no definite pattern of convergence but that sometimes converged monotonically, sometimes oscillated, or at times even diverged; also, convergence was not consistently either too stiff or too flexible. Melosh[5] presented a number of requirements with which the assumed displacement pattern over a discrete element must comply in order that monotonic convergence be assured; among these is the condition that adjacent elements must maintain displacement continuity along the common edge. Pian[6] pointed out, however, that, in order to achieve geometric admissibility for an assembled set of plate elements in bending, a slope continuity condition must also be imposed. A rectangular 16-degree-of-freedom plate element in bending for which the assumed displacement state satisfies the criteria of Melosh[5] as well as the interelement slope continuity requirement, and which does in fact exhibit monotonic convergence of the total potential energy, was introduced by Bogner, Fox, and Schmit.[7]

The major portion of the research pertaining to the development of finite elements for the modeling of shells has been devoted to closed rings or conical shell segments for the modeling of shells of revolution. A closed conical shell element for use within the framework of the displacement method was introduced by Grafton and Strome.[8] The use of the closed axisymmetric shell element is restricted in that only shells of revolution can be modeled and only special types of loadings can be handled. Jones and Strome[9] conclude that flat plate elements are important in the discrete element analysis of nonaxisymmetric shell structures. Fulton, Eppink, and Walz[10] show, however, that the bending behavior of a curved structure is not adequately represented by flat elements, and they suggest that the development of reliable curved elements would be a significant contribution to the finite-element method of structural analysis. Beitch[11] recognizes the need for curved elements; however, the displacement state of an assemblage of these elements is not geometrically admissable.

A finite cylindrical shell element that is capable of modeling a cylindrical shell subjected to general loadings and boundary conditions is reported herein. An assemblage composed of these elements possesses the important property that the deformed surface of the assemblage is geometrically admissible. Monotonic convergence of the total potential energy is therefore assured as the modeling is successively

Received May 25, 1966; revision received October 3, 1966. This research was supported by U. S. Air Force Contract AF 33(615)-3432. [6.02]

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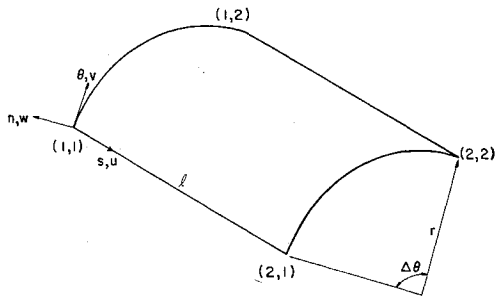


Fig. 1 Typical cylindrical element.

refined, since the assumed displacement pattern satisfies the requirements of admissible functions in the potential energy statement. Moreover, an eigenvalue analysis of a typical element stiffness matrix indicates that six independent rigid-body modes are adequately represented.

Displacement Modes

The assumed displacement state over the cylindrical shell element is expressed as products of one-dimensional, first-order Hermite interpolation polynomials using the lines of principal curvature as coordinates for a cylindrical surface (s, θ) (see Fig. 1). For the displacement in the longitudinal direction, we have

$$u(s,\theta) \approx \sum_{i=1}^{2}\sum_{j=1}^{2} [H_{0i}^{(1)}(s)\, H_{0j}^{(1)}(\theta)\, u_{ij} + H_{1i}^{(1)}(s) H_{0j}^{(1)}(\theta) \times u_{sij} + H_{0i}^{(1)}(s)\, H_{1j}^{(1)}(\theta) u_{\theta ij} + H_{1i}^{(1)}(s)\, H_{1j}^{(1)}(\theta)\, u_{s\theta ij}] \quad (1)$$

where, for example, $u_{ij} = u(s_i,\theta_j)$ are the nodal displacements, and $u_{\theta ij} = \partial u/r\partial\theta\ (s_i,\theta_j)$ are the nodal slopes in the circumferential direction. Expressions for the circumferential and normal displacements v and w are similar. The one-dimensional polynomials are given by

$$\left.\begin{aligned} H_{01}^{(1)}(s) &= (2s^3 - 3ls^2 + l^3)/l^3 \\ H_{02}^{(1)}(s) &= -(2s^3 - 3ls^2)/l^3 \\ H_{11}^{(1)}(s) &= (s^3 - 2ls^2 + l^2 s)/l^2 \\ H_{12}^{(1)}(s) &= (s^3 - ls^2)/l^2 \end{aligned}\right\} \quad (2)$$

Similar expressions for the θ direction are obtained by replacing s by $r\theta$, and l by $r\Delta\theta$. The properties of these polynomials and their graphs can be found in Ref. 7.

Potential Energy

The strain energy of an isotropic, linearly elastic, thin cylindrical shell element is given by

$$U = \int_0^l \int_0^{\Delta\theta} \int_{-h/2}^{h/2} \frac{E}{2(1-\nu^2)} \times \left[\epsilon_s^2 + \epsilon_\theta^2 + 2\nu\epsilon_s\epsilon_\theta + \frac{(1-\nu)}{2}\epsilon_{s\theta}^2\right] r\, ds d\theta dz \quad (3)$$

The linear strain displacement relations are written as

$$\left.\begin{aligned} \epsilon_s &= \partial u/\partial s - z\partial^2 w/\partial s^2 \\ \epsilon_\theta &= (\partial v/\partial\theta + w)/r - z(\partial^2 w/\partial\theta^2 - \partial v/\partial\theta)/r^2 \\ \epsilon_{s\theta} &= \partial v/\partial s + \partial u/r\partial\theta - 2z(\partial^2 w/\partial s\partial\theta - \partial v/\partial s)/r \end{aligned}\right\} \quad (4)$$

Table 1 Constants and exponents for symmetric submatrices

I	J	(1) G IJ	(2) G IJ	(3) G IJ	(4) G IJ	(5) G IJ	(6) G IJ	(7) G IJ	L IJ	M IJ	I	J	(1) G IJ	(2) G IJ	(3) G IJ	(4) G IJ	(5) G IJ	(6) G IJ	(7) G IJ	L IJ	M IJ
1	1	156/ 35	156/ 35	36/ 25	72/ 25	169/ 1225	78/ 175	78/ 175	0	0	12	3	-3/ 70	-13/210	-1/300	-1/150	13/ 58800	-1/ 1400	13/12600	1	2
2	1	78/ 35	22/ 35	3/ 25	36/ 25	143/ 7350	13/ 350	11/ 175	1	0	12	4	-1/ 70	-1/ 70	1/900	1/450	1/ 19600	1/ 4200	1/ 4200	2	2
2	2	52/ 35	4/ 35	4/ 25	8/ 25	13/ 3675	26/ 525	2/ 175	2	0	12	5	-11/ 35	13/ 70	-1/100	-3/ 25	143/ 88200	-11/ 2100	13/ 4200	1	1
3	1	22/ 35	78/ 35	3/ 25	36/ 25	143/ 7350	11/ 175	13/ 350	0	1	12	6	-11/105	3/ 70	1/300	1/ 25	11/ 29400	11/ 6300	1/ 1400	2	1
3	2	11/ 35	11/ 35	1/100	61/ 50	121/ 44100	11/ 2100	11/ 2100	1	1	12	7	2/ 35	-13/105	1/ 75	2/ 75	-13/ 44100	1/ 1050	-13/ 3150	1	2
3	3	4/ 35	52/ 35	4/ 25	8/ 25	13/ 3675	2/ 175	26/ 525	0	2	12	8	2/105	-1/ 35	-1/225	-2/225	-1/ 14700	-1/ 3150	-1/ 1050	2	2
4	1	11/ 35	11/ 35	1/100	11/ 50	121/ 44100	11/ 2100	11/ 2100	1	1	12	9	11/ 35	11/ 35	1/100	11/ 50	121/ 44100	11/ 2100	11/ 2100	1	1
4	2	22/105	2/ 35	1/ 75	4/ 25	11/ 22050	11/ 1575	1/ 1050	2	1	12	10	-22/105	-2/ 35	-1/ 75	-4/ 25	-11/ 22050	-11/ 1575	-1/ 1050	2	1
4	3	2/ 35	22/105	1/ 75	4/ 25	11/ 22050	1/ 1050	11/ 1575	1	2	12	11	-2/ 35	-22/105	-1/ 75	-4/ 25	-11/ 22050	-1/ 1050	-11/ 1575	1	2
4	4	4/105	4/105	4/225	8/225	1/ 11025	2/ 1575	2/ 1575	2	2	12	12	4/105	4/105	4/225	8/225	1/ 11025	2/ 1575	2/ 1575	2	2
5	1	54/ 35	-156/ 35	-36/ 25	-72/ 25	117/ 2450	27/ 175	-78/ 175	0	0	13	1	-156/ 35	54/ 35	-36/ 25	-72/ 25	117/ 2450	-78/ 175	27/ 175	0	0
5	2	27/ 35	-22/ 35	-3/ 25	-36/ 25	33/ 4900	9/ 700	-11/ 175	1	0	13	2	-78/ 35	13/ 35	-3/ 25	-6/ 25	169/ 14700	-13/ 350	13/ 350	1	0
5	3	13/ 35	-78/ 35	-3/ 25	-6/ 25	169/ 14700	13/ 350	-13/ 350	0	1	13	3	-22/ 35	27/ 35	-3/ 25	-36/ 25	33/ 4900	-11/ 175	9/ 700	0	1
5	4	13/ 70	-11/ 35	-1/100	-3/ 25	143/ 88200	13/ 4200	-11/ 2100	1	1	13	4	-11/ 35	13/ 70	-1/100	-3/ 25	143/ 88200	-11/ 2100	13/ 4200	1	1
5	5	156/ 35	156/ 35	36/ 25	72/ 25	169/ 1225	78/ 175	78/ 175	0	0	13	5	-54/ 35	-54/ 35	36/ 25	72/ 25	81/ 4900	-27/ 175	-27/ 175	0	0
6	1	27/ 35	-22/ 35	-3/ 25	-36/ 25	33/ 4900	9/ 700	-11/ 175	1	0	13	6	-27/ 35	-13/ 35	3/ 25	6/ 25	39/ 9800	-9/ 700	-13/ 350	1	0
6	2	18/ 35	-4/ 35	-4/ 25	-8/ 25	3/ 2450	3/ 175	-2/ 175	2	0	13	7	13/ 35	27/ 35	-3/ 25	-6/ 25	-39/ 9800	13/ 350	9/ 700	0	1
6	3	13/ 70	-11/ 35	-1/100	-3/ 25	143/ 88200	13/ 4200	-11/ 2100	1	1	13	8	13/ 70	13/ 70	-1/100	-1/ 50	-169/176400	13/ 4200	13/ 4200	1	1
6	4	13/105	-2/ 35	-1/ 75	-2/ 75	13/ 44100	13/ 3150	-1/ 1050	2	1	13	9	54/ 35	-156/ 35	-36/ 25	-72/ 25	117/ 2450	27/ 175	-78/ 175	0	0
6	5	78/ 35	22/ 35	3/ 25	36/ 25	143/ 7350	13/ 350	11/ 175	1	0	13	10	-27/ 35	22/ 35	3/ 25	36/ 25	-33/ 4900	-9/ 700	11/ 175	1	0
6	6	52/ 35	4/ 35	4/ 25	8/ 25	13/ 3675	26/ 525	2/ 175	2	0	13	11	-13/ 35	78/ 35	3/ 25	6/ 25	-169/ 14700	-13/ 350	13/ 350	0	1
7	1	-13/ 35	78/ 35	3/ 25	6/ 25	-169/ 14700	-13/ 350	13/ 350	0	1	13	12	13/ 70	-11/ 35	-1/100	-3/ 25	143/ 88200	13/ 4200	-11/ 2100	1	1
7	2	-13/ 70	11/ 35	1/100	3/ 25	-143/ 88200	-13/ 4200	11/ 2100	1	1	13	13	156/ 35	156/ 35	36/ 25	72/ 25	169/ 1225	78/ 175	78/ 175	0	0
7	3	-3/ 35	26/ 35	-1/ 25	-2/ 25	-13/ 4900	-3/ 350	-13/ 1050	0	2	14	1	78/ 35	-13/ 35	3/ 25	6/ 25	-169/ 14700	13/ 350	-13/ 350	1	0
7	4	-3/ 70	11/105	-1/300	-1/ 25	-11/ 29400	-1/ 1400	-11/ 6300	1	2	14	2	26/ 35	-3/ 35	-1/ 25	-2/ 25	-13/ 4900	-13/ 1050	-3/ 350	2	0
7	5	-22/ 35	-78/ 35	-3/ 25	-36/ 25	-143/ 7350	-11/ 175	-13/ 350	0	1	14	3	11/ 35	-13/ 70	1/100	3/ 25	-143/ 88200	11/ 2100	-13/ 4200	1	1
7	6	-11/ 35	-11/ 35	-1/100	-61/ 50	-121/ 44100	-11/ 2100	-11/ 2100	1	1	14	4	11/105	-3/ 70	-1/300	-1/ 25	-11/ 29400	-11/ 6300	-1/ 1400	2	1
7	7	4/ 35	52/ 35	4/ 25	8/ 25	13/ 3675	2/ 175	26/ 525	0	2	14	5	27/ 35	13/ 35	-3/ 25	-6/ 25	-39/ 9800	9/ 700	13/ 350	1	0
8	1	-13/ 70	11/ 35	1/100	3/ 25	-143/ 88200	-13/ 4200	11/ 2100	1	1	14	6	9/ 35	3/ 35	1/ 25	2/ 25	-9/ 9800	-3/ 700	3/ 350	2	0
8	2	-13/105	2/ 35	1/ 75	2/ 75	-13/ 44100	-13/ 3150	1/ 1050	2	1	14	7	-13/ 70	-13/ 70	1/100	1/ 50	169/176400	-13/ 4200	-13/ 4200	1	1
8	3	-3/ 70	11/105	-1/300	-1/ 25	-11/ 29400	-1/ 1400	-11/ 6300	1	2	14	8	-13/210	-3/ 70	-1/300	-1/150	13/ 58800	13/12600	-1/ 1400	2	1
8	4	-1/ 35	2/105	-1/225	-2/225	-1/ 14700	-1/ 1050	-1/ 3150	2	2	14	9	-27/ 35	22/ 35	3/ 25	36/ 25	-33/ 4900	-9/ 700	11/ 175	1	0
8	5	-11/ 35	-11/ 35	-1/100	-11/ 50	-121/ 44100	-11/ 2100	-11/ 2100	1	1	14	10	18/ 35	-4/ 35	-4/ 25	-8/ 25	3/ 2450	3/ 175	-2/ 175	2	0
8	6	-22/105	-2/ 35	-1/ 75	-4/ 25	-11/ 22050	-11/ 1575	-1/ 1050	2	1	14	11	13/ 70	-11/ 35	-1/100	-3/ 25	143/ 88200	13/ 4200	-11/ 2100	1	1
8	7	2/ 35	22/105	1/ 75	4/ 25	11/ 22050	1/ 1050	11/ 1575	1	2	14	12	-13/105	2/ 35	1/ 75	2/ 75	-13/ 44100	-13/ 3150	1/ 1050	2	1
8	8	4/105	4/105	4/225	8/225	1/ 11025	2/ 1575	2/ 1575	2	2	14	13	-78/ 35	-22/ 35	-3/ 25	-36/ 25	-143/ 7350	-13/ 350	-11/ 175	1	0
9	1	-54/ 35	-54/ 35	36/ 25	72/ 25	81/ 4900	-27/ 175	-27/ 175	0	0	14	14	52/ 35	4/ 35	4/ 25	8/ 25	13/ 3675	26/ 525	2/ 175	2	0
9	2	-27/ 35	-13/ 35	3/ 25	6/ 25	39/ 9800	-9/ 700	-13/ 350	1	0	15	1	-22/ 35	27/ 35	-3/ 25	-36/ 25	33/ 4900	-11/ 175	9/ 700	0	1
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9	4	-13/ 70	-13/ 70	1/100	1/ 50	169/176400	-13/ 4200	-13/ 4200	1	1	15	3	-4/ 35	18/ 35	-4/ 25	-8/ 25	3/ 2450	-2/ 175	3/ 175	0	2
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9	7	22/ 35	-27/ 35	3/ 25	36/ 25	-33/ 4900	11/ 175	-9/ 700	0	1	15	6	-13/ 70	13/ 70	1/100	1/ 50	169/176400	-13/ 4200	-13/ 4200	1	1
9	8	11/ 35	-13/ 70	1/100	3/ 25	-143/ 88200	11/ 2100	-13/ 4200	1	1	15	7	3/ 35	9/ 35	1/ 25	2/ 25	-9/ 9800	3/ 350	-3/ 700	0	2
9	9	156/ 35	156/ 35	36/ 25	72/ 25	169/ 1225	78/ 175	78/ 175	0	0	15	8	3/ 70	13/210	1/300	1/150	-13/ 58800	1/ 1400	-13/12600	1	2
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10	2	9/ 35	3/ 35	1/ 25	2/ 25	-9/ 9800	-3/ 700	3/ 350	2	0	15	10	-13/ 70	11/ 35	1/100	3/ 25	-143/ 88200	-13/ 4200	11/ 2100	1	1
10	3	13/ 70	13/ 70	-1/100	-1/ 50	-169/176400	13/ 4200	13/ 4200	1	1	15	11	-3/ 35	26/ 35	-1/ 25	-2/ 25	-13/ 4900	-3/ 350	-13/ 1050	0	2
10	4	13/210	3/ 70	1/300	1/150	-13/ 58800	-13/12600	1/ 1400	2	1	15	12	3/ 70	-11/105	1/300	1/ 25	11/ 29400	1/ 1400	11/ 6300	1	2
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10	7	-11/ 35	13/ 70	-1/100	-3/ 25	143/ 88200	-11/ 2100	13/ 4200	1	1	15	15	4/ 35	52/ 35	4/ 25	8/ 25	13/ 3675	2/ 175	26/ 525	0	2
10	8	-11/105	3/ 70	1/300	1/ 25	11/ 29400	11/ 6300	1/ 1400	2	1	16	1	11/ 35	-13/ 70	1/100	3/ 25	-143/ 88200	11/ 2100	-13/ 4200	1	1
10	9	-78/ 35	-22/ 35	-3/ 25	-36/ 25	-143/ 7350	-13/ 350	-11/ 175	1	0	16	2	11/105	-3/ 70	-1/300	-1/ 25	-11/ 29400	-11/ 6300	-1/ 1400	2	1
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11	3	3/ 35	9/ 35	1/ 25	2/ 25	-9/ 9800	3/ 350	-3/ 700	0	2	16	6	13/210	3/ 70	1/300	1/150	-13/ 58800	-13/12600	1/ 1400	2	1
11	4	3/ 70	13/210	1/300	1/150	-13/ 58800	1/ 1400	-13/12600	1	2	16	7	-3/ 70	-13/210	-1/300	-1/150	13/ 58800	-1/ 1400	13/12600	1	2
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11	7	-4/ 35	18/ 35	-4/ 25	-8/ 25	3/ 2450	-2/ 175	3/ 175	0	2	16	10	13/105	-2/ 35	-1/ 75	-2/ 75	13/ 44100	13/ 3150	-1/ 1050	2	1
11	8	-2/ 35	13/105	-1/ 75	-2/ 75	13/ 44100	-1/ 1050	13/ 3150	1	2	16	11	3/ 70	-11/105	1/300	1/ 25	11/ 29400	1/ 1400	11/ 6300	1	2
11	9	-22/ 35	-78/ 35	-3/ 25	-36/ 25	-143/ 7350	-11/ 175	-13/ 350	0	1	16	12	-1/ 35	2/105	-1/225	-2/225	-1/ 14700	-1/ 1050	-1/ 3150	2	2
11	10	11/ 35	11/ 35	1/100	61/ 50	121/ 44100	11/ 2100	11/ 2100	1	1	16	13	-11/ 35	-11/ 35	-1/100	-11/ 50	-121/ 44100	-11/ 2100	-11/ 2100	1	1
11	11	4/ 35	52/ 35	4/ 25	8/ 25	13/ 3675	2/ 175	26/ 525	0	2	16	14	22/105	2/ 35	1/ 75	4/ 25	11/ 22050	11/ 1575	1/ 1050	2	1
12	1	-13/ 70	-13/ 70	1/100	1/ 50	169/176400	-13/ 4200	-13/ 4200	1	1	16	15	-2/ 35	-22/105	-1/ 75	-4/ 25	-11/ 22050	-1/ 1050	-11/ 1575	1	2
12	2	-13/210	-3/ 70	-1/300	-1/150	13/ 58800	13/12600	-1/ 1400	2	1	16	16	4/105	4/105	4/225	8/225	1/ 11025	2/ 1575	2/ 1575	2	2

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Upon substitution of Eqs. (4) and performing the integration with respect to z, the strain energy [Eq. (3)] becomes

$$U = \frac{D}{2}\int_0^l\int_0^{\Delta\theta}\left\{\left(\frac{\partial^2 w}{\partial s^2}\right)^2 + \frac{1}{r^4}\left(\frac{\partial^2 w}{\partial\theta^2}\right)^2 + \frac{2\nu}{r^2}\frac{\partial^2 w}{\partial s^2}\frac{\partial^2 w}{\partial\theta^2} + \frac{2(1-\nu)}{r^2}\left(\frac{\partial^2 w}{\partial s\partial\theta}\right)^2 + \frac{12}{h^2r^2}w^2 + \frac{12}{h^2}\left[\left(\frac{\partial u}{\partial s}\right)^2 + \frac{1-\nu}{2r^2}\left(\frac{\partial u}{\partial\theta}\right)^2\right] + \frac{1}{r^2}\left(\frac{12}{h^2}+\frac{1}{r^2}\right)\times\left(\frac{\partial v}{\partial\theta}\right)^2 + 2(1-\nu)\left(\frac{3}{h^2}+\frac{1}{r^2}\right)\left(\frac{\partial v}{\partial s}\right)^2 + \frac{24\nu}{h^2r}w\frac{\partial u}{\partial s} + \frac{24}{h^2r^2}w\frac{\partial v}{\partial\theta} - \frac{2}{r^4}\frac{\partial^2 w}{\partial\theta^2}\frac{\partial v}{\partial\theta} - \frac{4(1-\nu)}{r^2}\frac{\partial^2 w}{\partial s\partial\theta}\frac{\partial v}{\partial s} - \frac{2\nu}{r^2}\frac{\partial^2 w}{\partial s^2}\frac{\partial v}{\partial\theta} + \frac{12}{h^2}\left[\frac{2\nu}{r}\frac{\partial u}{\partial s}\frac{\partial v}{\partial\theta} + \frac{1-\nu}{r}\frac{\partial u}{\partial\theta}\frac{\partial v}{\partial s}\right]\right\} rds\,d\theta \quad (5)$$

where $D = Eh^3/12(1-\nu^2)$, the flexural rigidity. The external work term W is written as

$$W = \int_0^l\int_0^{\Delta\theta}[p_u u + p_v v + p_w w]rds\,d\theta \quad (6)$$

where $p_u(s,\theta)$, $p_v(s,\theta)$, and $p_w(s,\theta)$ denote middle surface tractions in the u, v, and w directions, respectively.

The total potential energy $\Pi_p^{(k)}$ for the kth cylindrical shell element is given by

$$\Pi_p^{(k)} = U^{(k)} - W^{(k)} \quad (7)$$

The total potential energy of an assembly of N cylindrical elements is

$$\Pi_p = \sum_{k=1}^{N}\Pi_p^{(k)} \quad (8)$$

where $\Pi_p^{(k)}$ is the potential energy of the kth element as computed from Eq. (7).

Element Stiffness Matrix

When the assumed displacement modes [Eq. (1)] are substituted into the expression characterizing the strain energy [Eq. (5)] and the indicated integration is carried out over the middle surface of the finite cylindrical shell element depicted in Fig. 1, then the strain energy of the element is given by

$$U = \tfrac{1}{2}X^T K X \quad (9)$$

where X is a vector containing the 48 nodal variables of the element:

$$X^T = [u_{11}, u_{s11}, u_{\theta 11}, u_{s\theta 11}, u_{12}, \ldots, u_{s\theta 12}, u_{22}, \ldots, u_{s\theta 22}, u_{21}, \ldots, u_{s\theta 21}, v_{11}, v_{s11}, v_{\theta 11}, v_{s\theta 11}, v_{12}, \ldots, v_{s\theta 12}, v_{22}, \ldots, v_{s\theta 22}, v_{21}, \ldots, v_{s\theta 21}, w_{11}, w_{s11}, w_{\theta 11}, w_{s\theta 11}, w_{12}, \ldots, w_{s\theta 12}, w_{22}, \ldots, w_{s\theta 22}, w_{21}, \ldots, w_{s\theta 21}] \quad (10)$$

The element stiffness matrix K, which has dimensions of 48 × 48, is partioned as follows:

$$K = \left[\begin{array}{c|c|c} K^{(u)} & K^{(vu)T} & K^{(wu)T} \\ \hline K^{(vu)} & K^{(v)} & K^{(wv)T} \\ \hline K^{(wu)} & K^{(wv)} & K^{(w)} \end{array}\right] \quad (11)$$

where all of the submatrices are 16 × 16 dimensional arrays. The submatrices on the diagonal are symmetric and contain contributions that do not involve coupling between the u, v, and w displacements. The off-diagonal arrays contain contributions from the coupling terms as indicated by the superscripts and are not symmetric; the total element stiffness matrix K is, however, symmetric.

The elements of the submatrices are given by the following formulas, where the constants $G_{ij}^{(k)}$, $k = 1, 2, \ldots, 14$, and the exponents L_{ij} and M_{ij} are given in Tables 1 and 2:

$$\left.\begin{aligned}
K_{ij}^{(u)} &= \frac{D}{lr\Delta\theta}\frac{12}{h^2}\left[(r\Delta\theta)^2 G_{ij}^{(6)} + \frac{(1-\nu)l^2}{2}G_{ij}^{(7)}\right] l^{L_{ij}}(r\Delta\theta)^{M_{ij}} \\
K_{ij}^{(v)} &= \frac{D}{lr\Delta\theta}\left[\left(\frac{12}{h^2}+\frac{1}{r^2}\right)l^2 G_{ij}^{(7)} + 2(1-\nu)\times\left(\frac{3}{h^2}+\frac{1}{r^2}\right)(r\Delta\theta)^2 G_{ij}^{(6)}\right] l^{L_{ij}}(r\Delta\theta)^{M_{ij}} \\
K_{ij}^{(w)} &= \frac{D}{lr\Delta\theta}\left[\left(\frac{r\Delta\theta}{l}\right)^2 G_{ij}^{(1)} + \left(\frac{l}{r\Delta\theta}\right)^2 G_{ij}^{(2)} + \nu G_{ij}^{(4)} + 2(1-\nu)G_{ij}^{(3)} + \frac{12l^2\Delta\theta^2}{h^2}G_{ij}^{(5)}\right]\times l^{L_{ij}}(r\Delta\theta)^{M_{ij}} \\
K_{ij}^{(vu)} &= \frac{D}{lr\Delta\theta}\frac{12}{h^2}\left[\nu\, l\, r\Delta\theta G_{ij}^{(9)} + \left(\frac{1-\nu}{2}\right) l\, r\Delta\theta G_{ij}^{(8)}\right] l^{L_{ij}}(r\Delta\theta)^{M_{ij}} \\
K_{ij}^{(wu)} &= \frac{D}{lr\Delta\theta}\frac{12}{h^2}\left[\nu\, l\, r\Delta\theta^2 G_{ij}^{(10)}\right] l^{L_{ij}}(r\Delta\theta)^{M_{ij}} \\
K_{ij}^{(wv)} &= \frac{D}{lr\Delta\theta}\frac{12\,l^2\Delta\theta}{h^2}\left[G_{ij}^{(11)} - \frac{l^2}{r^2\Delta\theta}G_{ij}^{(12)} - 2(1-\nu)\Delta\theta\, G_{ij}^{(13)} - \nu\Delta\theta\, G_{ij}^{(14)}\right] l^{L_{ij}}(r\Delta\theta)^{M_{ij}}
\end{aligned}\right\} \quad (12)$$

Element Mass Matrix

The kinetic energy for a cylindrical element takes the following form after integration over the thickness:

$$T = \frac{1}{2}\int_0^l\int_0^{\Delta\theta}\rho h\left\{\left(\frac{\partial u}{\partial t}\right)^2 + \left(\frac{\partial v}{\partial t}\right)^2 + \left(\frac{\partial w}{\partial t}\right)^2 + h^2\left[\frac{(\partial^2 w/\partial s\partial t)^2 + (\partial^2 w/\partial\theta\partial t)^2/r^2}{12}\right]\right\} rds\,d\theta \quad (13)$$

where the last two terms represent the effects of rotary inertia.§ If the usual assumption that the displacements are sinusoidal in time is made, then the term $(\partial w/\partial t)^2$ can be replaced by $\omega^2 w^2$, and similarly with the other terms. Then Eq. (13) becomes

$$T = \frac{1}{2}\int_0^l\int_0^{\Delta\theta}\rho h\,\omega^2\left\{u^2 + v^2 + w^2 + h^2\left[\frac{(\partial w/\partial s)^2 + (\partial w/\partial\theta)^2/r^2}{12}\right]\right\} rds\,d\theta \quad (14)$$

Substituting the displacement modes [Eq. (1)] into Eq. (14) and carrying out the integration over the middle surface of the finite element, the kinetic energy can be written as

$$T = (\omega^2/2)X^T M X \quad (15)$$

where X is the vector of nodal variables [Eq. (10)]. Forming the Lagrangian function $L = T - U$ and setting the variation to zero, the following standard eigenvalue problem is obtained:

$$K X = \omega^2 M X \quad (16)$$

where K is the element stiffness matrix [Eq. (11)] and M is the element mass matrix consistent with the assumed dis-

§ The rotary inertia terms were obtained by assuming that $z/r \ll 1$ in the circumferential displacement expression off the middle surface.

Table 2 Constants and exponents for unsymmetric submatrices

I	J	(8) G IJ	(9) G IJ	(10) G IJ	(11) G IJ	(12) G IJ	(13) G IJ	(14) G IJ	L IJ	M IJ
1	1	1/ 4	1/ 4	-13/ 70	-13/ 70	0/ 1	-3/ 5	3/ 5	0	0
1	2	1/ 20	-1/ 20	13/ 350	-11/ 420	0/ 1	-1/ 20	1/ 20	1	0
1	3	-1/ 20	1/ 20	-11/ 420	13/ 350	-13/ 35	-3/ 25	-3/ 25	0	1
1	4	-1/ 100	-1/ 100	11/ 2100	11/ 2100	-11/210	-1/ 100	-1/ 100	1	1
1	5	-1/ 4	1/ 4	-9/ 140	13/ 70	0/ 1	-3/ 5	-3/ 5	0	0
1	6	-1/ 20	-1/ 20	9/ 700	11/ 420	0/ 1	-1/ 20	-1/ 20	1	0
1	7	1/ 20	-1/ 20	13/ 840	-13/ 350	13/ 35	3/ 25	3/ 25	0	1
1	8	1/ 100	1/ 100	-13/ 4200	-11/ 2100	11/210	1/ 100	1/ 100	1	1
1	9	-1/ 4	-1/ 4	9/ 140	9/ 140	0/ 1	3/ 5	3/ 5	0	0
1	10	1/ 20	1/ 20	-9/ 700	-13/ 840	-0/ 1	-1/ 20	-1/ 20	1	0
1	11	1/ 20	1/ 20	-13/ 840	-9/ 700	9/ 70	-3/ 25	-3/ 25	0	1
1	12	-1/ 100	-1/ 100	13/ 4200	13/ 4200	-13/420	1/ 100	1/ 100	1	1
1	13	1/ 4	-1/ 4	13/ 70	-9/ 140	0/ 1	3/ 5	-3/ 5	0	0
1	14	-1/ 20	1/ 20	-13/ 350	13/ 840	-0/ 1	-1/ 20	1/ 20	1	0
1	15	-1/ 20	-1/ 20	11/ 420	9/ 700	-9/ 70	3/ 25	3/ 25	0	1
1	16	1/ 100	1/ 100	-11/ 2100	-13/ 4200	13/420	-1/ 100	-1/ 100	1	1
2	1	-1/ 20	1/ 20	-13/ 350	-11/ 420	0/ 1	-1/ 20	11/ 20	1	0
2	2	-0/ 1	-0/ 1	0/ 1	-1/ 210	0/ 1	-1/ 15	1/ 15	2	0
2	3	1/ 100	1/ 100	-11/ 2100	11/ 2100	-11/210	-1/ 100	-11/ 100	1	1
2	4	0/ 1	-0/ 1	0/ 1	1/ 1050	-1/105	-1/ 75	-1/ 75	2	1
2	5	1/ 20	1/ 20	-9/ 700	11/ 420	0/ 1	-1/ 20	-11/ 20	1	0
2	6	0/ 1	-0/ 1	0/ 1	1/ 210	0/ 1	-1/ 15	-1/ 15	2	0
2	7	-1/ 100	-1/ 100	13/ 4200	-11/ 2100	11/210	1/ 100	11/ 100	1	1
2	8	-0/ 1	0/ 1	-0/ 1	-1/ 1050	1/105	1/ 75	1/ 75	2	1
2	9	-1/ 20	-1/ 20	9/ 700	13/ 840	0/ 1	1/ 20	1/ 20	1	0
2	10	1/ 120	1/ 120	-3/ 1400	-1/ 280	-0/ 1	1/ 60	1/ 60	2	0
2	11	1/ 100	1/ 100	-13/ 4200	-13/ 4200	13/420	-1/ 100	-1/ 100	1	1
2	12	-1/ 600	-1/ 600	13/25200	1/ 1400	-1/140	-1/ 300	-1/ 300	2	1
2	13	1/ 20	-1/ 20	13/ 350	-13/ 840	0/ 1	1/ 20	-1/ 20	1	0
2	14	-1/ 120	1/ 120	-13/ 2100	1/ 280	-0/ 1	1/ 60	-1/ 60	2	0
2	15	-1/ 100	-1/ 100	11/ 2100	13/ 4200	-13/420	1/ 100	1/ 100	1	1
2	16	1/ 600	1/ 600	-11/12600	-1/ 1400	1/140	1/ 300	1/ 300	2	1
3	1	1/ 20	-1/ 20	-11/ 420	-13/ 350	13/ 35	3/ 25	3/ 25	0	1
3	2	1/ 100	1/ 100	11/ 2100	-11/ 2100	11/210	1/ 100	1/ 100	1	1
3	3	-0/ 1	-0/ 1	-1/ 210	0/ 1	-13/ 70	0/ 1	-0/ 1	0	2
3	4	-0/ 1	0/ 1	1/ 1050	0/ 1	-11/420	0/ 1	-0/ 1	1	2
3	5	-1/ 20	1/ 20	-13/ 840	13/ 350	-13/ 35	-3/ 25	-3/ 25	0	1
3	6	-1/ 100	-1/ 100	13/ 4200	11/ 2100	-11/210	-1/ 100	-1/ 100	1	1
3	7	1/ 120	-1/ 120	1/ 280	-13/ 2100	13/ 70	1/ 50	1/ 50	0	2
3	8	1/ 600	1/ 600	-1/ 1400	-11/12600	11/420	1/ 600	1/ 600	1	2
3	9	-1/ 20	-1/ 20	13/ 840	9/ 700	-9/ 70	3/ 25	3/ 25	0	1
3	10	1/ 100	1/ 100	-13/ 4200	-13/ 4200	13/420	-1/ 100	-1/ 100	1	1
3	11	1/ 120	1/ 120	-1/ 280	-3/ 1400	9/140	-1/ 50	1/ 50	0	2
3	12	-1/ 600	-1/ 600	1/ 1400	13/25200	-13/840	1/ 600	1/ 600	1	2
3	13	1/ 20	1/ 20	11/ 420	-9/ 700	9/ 70	-3/ 25	-3/ 25	0	1
3	14	-1/ 100	-1/ 100	-11/ 2100	13/ 4200	-13/420	1/ 100	1/ 100	1	1
3	15	-0/ 1	0/ 1	1/ 210	0/ 1	-9/140	-0/ 1	0/ 1	0	2
3	16	0/ 1	-0/ 1	-1/ 1050	-0/ 1	13/840	0/ 1	-0/ 1	1	2
4	1	-1/ 100	-1/ 100	-11/ 2100	-11/ 2100	11/210	1/ 100	11/ 100	1	1
4	2	-0/ 1	0/ 1	0/ 1	-1/ 1050	1/105	1/ 75	1/ 75	2	1
4	3	0/ 1	-0/ 1	-1/ 1050	0/ 1	-11/420	0/ 1	-0/ 1	1	2
4	4	0/ 1	0/ 1	0/ 1	0/ 1	-1/210	0/ 1	-0/ 1	2	2
4	5	1/ 100	1/ 100	-13/ 4200	11/ 2100	-11/210	-1/ 100	-11/ 100	1	1
4	6	0/ 1	-0/ 1	0/ 1	1/ 1050	-1/105	-1/ 75	-1/ 75	2	1
4	7	-1/ 600	-1/ 600	1/ 1400	-11/12600	11/420	1/ 600	11/ 600	1	2
4	8	-0/ 1	0/ 1	-0/ 1	-1/ 6300	1/210	1/ 450	1/ 450	2	2
4	9	-1/ 100	-1/ 100	13/ 4200	13/ 4200	-13/420	1/ 100	1/ 100	1	1
4	10	1/ 600	1/ 600	-13/25200	-1/ 1400	1/140	1/ 300	1/ 300	2	1
4	11	1/ 600	1/ 600	-1/ 1400	-13/25200	13/840	-1/ 600	-1/ 600	1	2
4	12	-1/3600	-1/3600	1/ 8400	1/ 8400	-1/280	-1/1800	-1/1800	2	2
4	13	1/ 100	1/ 100	11/ 2100	-13/ 4200	13/420	-1/ 100	-1/ 100	1	1
4	14	-1/ 600	-1/ 600	-11/12600	1/ 1400	-1/140	-1/ 300	-1/ 300	2	1
4	15	-0/ 1	0/ 1	1/ 1050	0/ 1	-13/840	-0/ 1	0/ 1	1	2
4	16	0/ 1	-0/ 1	-1/ 6300	-0/ 1	1/280	-0/ 1	0/ 1	2	2
5	1	1/ 4	-1/ 4	-9/ 140	-13/ 70	0/ 1	3/ 5	3/ 5	0	0
5	2	1/ 20	1/ 20	9/ 700	-11/ 420	0/ 1	1/ 20	1/ 20	1	0
5	3	1/ 20	-1/ 20	-13/ 840	13/ 350	13/ 35	3/ 25	3/ 25	0	1
5	4	1/ 100	1/ 100	13/ 4200	-11/ 2100	11/210	1/ 100	1/ 100	1	1
5	5	-1/ 4	-1/ 4	-13/ 70	13/ 70	0/ 1	3/ 5	-3/ 5	0	0
5	6	-1/ 20	1/ 20	13/ 350	11/ 420	0/ 1	1/ 20	-1/ 20	1	0
5	7	-1/ 20	1/ 20	11/ 420	13/ 350	-13/ 35	-3/ 25	-3/ 25	0	1
5	8	-1/ 100	-1/ 100	-11/ 2100	11/ 2100	-11/210	-1/ 100	-1/ 100	1	1
5	9	-1/ 4	1/ 4	13/ 70	-9/ 140	0/ 1	-3/ 5	3/ 5	0	0
5	10	1/ 20	-1/ 20	-13/ 350	-13/ 840	-0/ 1	1/ 20	-1/ 20	1	0
5	11	-1/ 20	-1/ 20	-11/ 420	-9/ 700	-9/ 70	3/ 25	3/ 25	0	1
5	12	1/ 100	1/ 100	11/ 2100	-13/ 4200	13/420	-1/ 100	-1/ 100	1	1
5	13	1/ 4	1/ 4	-9/ 140	-9/ 140	0/ 1	-3/ 5	-3/ 5	0	0
5	14	-1/ 20	-1/ 20	-9/ 700	13/ 840	-0/ 1	1/ 20	1/ 20	1	0
5	15	1/ 20	1/ 20	13/ 840	-9/ 700	9/ 70	-3/ 25	-3/ 25	0	1
5	16	-1/ 100	-1/ 100	-13/ 4200	13/ 4200	-13/420	1/ 100	1/ 100	1	1
6	1	-1/ 20	-1/ 20	-9/ 700	-11/ 420	0/ 1	1/ 20	11/ 20	1	0
6	2	-0/ 1	0/ 1	0/ 1	-1/ 210	0/ 1	1/ 15	1/ 15	2	0
6	3	-1/ 100	-1/ 100	-13/ 4200	-11/ 2100	11/210	-1/ 100	11/ 100	1	1
6	4	-0/ 1	0/ 1	0/ 1	-1/ 1050	1/105	1/ 75	1/ 75	2	1
6	5	1/ 20	-1/ 20	-13/ 350	11/ 420	0/ 1	1/ 20	-11/ 20	1	0
6	6	0/ 1	0/ 1	0/ 1	1/ 210	0/ 1	1/ 15	-1/ 15	2	0
6	7	1/ 100	1/ 100	11/ 2100	11/ 2100	-11/210	-1/ 100	-11/ 100	1	1
6	8	0/ 1	-0/ 1	-0/ 1	1/ 1050	-1/105	-1/ 75	-1/ 75	2	1
6	9	-1/ 20	1/ 20	13/ 350	13/ 840	0/ 1	-1/ 20	1/ 20	1	0
6	10	1/ 120	-1/ 120	-13/ 2100	-1/ 280	-0/ 1	-1/ 60	1/ 60	2	0
6	11	-1/ 100	-1/ 100	-11/ 2100	13/ 4200	-13/420	1/ 100	1/ 100	1	1
6	12	1/ 600	1/ 600	11/12600	-1/ 1400	1/140	1/ 300	1/ 300	2	1
6	13	1/ 20	1/ 20	9/ 700	-13/ 840	0/ 1	-1/ 20	-1/ 20	1	0
6	14	-1/ 120	-1/ 120	-3/ 1400	1/ 280	-0/ 1	-1/ 60	-1/ 60	2	0
6	15	1/ 100	1/ 100	13/ 4200	-13/ 4200	13/420	-1/ 100	-1/ 100	1	1
6	16	-1/ 600	-1/ 600	-13/25200	1/ 1400	-1/140	-1/ 300	-1/ 300	2	1
7	1	-1/ 20	1/ 20	13/ 840	13/ 350	-13/ 35	-3/ 25	-3/ 25	0	1
7	2	-1/ 100	-1/ 100	-13/ 4200	11/ 2100	-11/210	-1/ 100	-1/ 100	1	1
7	3	-1/ 120	1/ 120	1/ 280	13/ 2100	-13/ 70	-1/ 50	-1/ 50	0	2
7	4	-1/ 600	-1/ 600	-1/ 1400	11/12600	-11/420	-1/ 600	-1/ 600	1	2
7	5	1/ 20	-1/ 20	11/ 420	-13/ 350	13/ 35	3/ 25	3/ 25	0	1
7	6	1/ 100	1/ 100	-11/ 2100	-11/ 2100	11/210	1/ 100	1/ 100	1	1
7	7	-0/ 1	-0/ 1	-1/ 210	0/ 1	13/ 70	0/ 1	-0/ 1	0	2
7	8	-0/ 1	0/ 1	1/ 1050	0/ 1	11/420	0/ 1	-0/ 1	1	2
7	9	1/ 20	1/ 20	-11/ 420	-9/ 700	9/ 70	-3/ 25	-3/ 25	0	1
7	10	-1/ 100	-1/ 100	11/ 2100	13/ 4200	-13/420	1/ 100	1/ 100	1	1
7	11	-0/ 1	0/ 1	1/ 210	0/ 1	9/140	-0/ 1	0/ 1	0	2
7	12	0/ 1	-0/ 1	-1/ 1050	-0/ 1	-13/840	0/ 1	-0/ 1	1	2
7	13	-1/ 20	-1/ 20	-13/ 840	9/ 700	-9/ 70	3/ 25	3/ 25	0	1
7	14	1/ 100	1/ 100	13/ 4200	-13/ 4200	13/420	-1/ 100	-1/ 100	1	1
7	15	-1/ 120	-1/ 120	-1/ 280	3/ 1400	-9/140	1/ 50	1/ 50	0	2
7	16	1/ 600	1/ 600	1/ 1400	-13/25200	13/840	-1/ 600	-1/ 600	1	2
8	1	1/ 100	1/ 100	13/ 4200	11/ 2100	-11/210	-1/ 100	-11/ 100	1	1
8	2	0/ 1	-0/ 1	-0/ 1	1/ 1050	-1/105	-1/ 75	-1/ 75	2	1
8	3	1/ 600	1/ 600	1/ 1400	11/12600	-11/420	-1/ 600	-11/ 600	1	2
8	4	0/ 1	-0/ 1	-0/ 1	1/ 6300	-1/210	-1/ 450	-1/ 450	2	2
8	5	-1/ 100	-1/ 100	11/ 2100	11/ 2100	11/210	1/ 100	11/ 100	1	1
8	6	-0/ 1	0/ 1	-0/ 1	-1/ 1050	1/105	1/ 75	1/ 75	2	1
8	7	0/ 1	-0/ 1	-1/ 1050	0/ 1	11/420	0/ 1	-0/ 1	1	2
8	8	0/ 1	0/ 1	0/ 1	0/ 1	1/210	0/ 1	-0/ 1	2	2
8	9	1/ 100	1/ 100	-11/ 2100	-13/ 4200	13/420	-1/ 100	-1/ 100	1	1
8	10	-1/ 600	-1/ 600	11/12600	1/ 1400	-1/140	-1/ 300	-1/ 300	2	1
8	11	-0/ 1	0/ 1	1/ 1050	0/ 1	13/840	-0/ 1	0/ 1	1	2
8	12	0/ 1	-0/ 1	-1/ 6300	-0/ 1	-1/280	-0/ 1	0/ 1	2	2
8	13	-1/ 100	-1/ 100	-13/ 4200	13/ 4200	-13/420	1/ 100	1/ 100	1	1
8	14	1/ 600	1/ 600	13/25200	-1/ 1400	1/140	1/ 300	1/ 300	2	1
8	15	-1/ 600	-1/ 600	-1/ 1400	13/25200	-13/840	1/ 600	1/ 600	1	2
8	16	1/3600	1/3600	1/ 8400	-1/ 8400	1/280	1/1800	1/1800	2	2

placement state. M is partioned as follows:

$$M = \begin{bmatrix} M^{(u)} & 0 & 0 \\ 0 & M^{(v)} & 0 \\ 0 & 0 & M^{(w)} \end{bmatrix} \tag{17}$$

The elements of the symmetric 16 × 16 dimensional submatrices are given by

$$M_{ij}^{(u)} = M_{ij}^{(v)} = \rho h\, l\, r \Delta\theta\, G_{ij}^{(5)}\, l^{L_{ij}} (r\Delta\theta)^{M_{ij}}$$

$$M_{ij}^{(w)} = \frac{\rho h\, lr\, \Delta\theta}{12} \left[12\, G_{ij}^{(5)} + \left(\frac{h}{l}\right)^2 G_{ij}^{(6)} + \left(\frac{h}{r\Delta\theta}\right)^2 G_{ij}^{(7)} \right] l^{L_{ij}} (r\Delta\theta)^{M_{ij}} \tag{18}$$

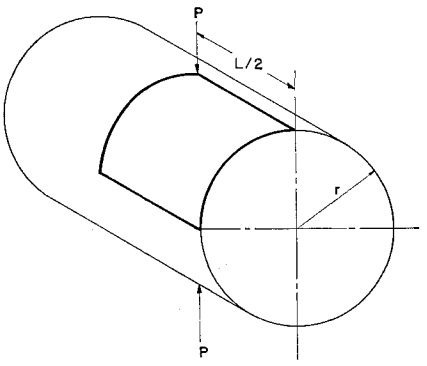


Fig. 2 Pinched cylinder (P = 100 lb, L = 10.35 in., r = 4.953 in., h = 0.094 in., ν = 0.3125, E = 10.5 × 10^6 psi).

where the constants $G_{ij}^{(k)}$, k = 5, 6, 7, and the exponents L_{ij} and M_{ij} are found in Table 1.

Numerical Example

To demonstrate the effectiveness of the proposed cylindrical element, the pinched cylinder shown in Fig. 2 is con-

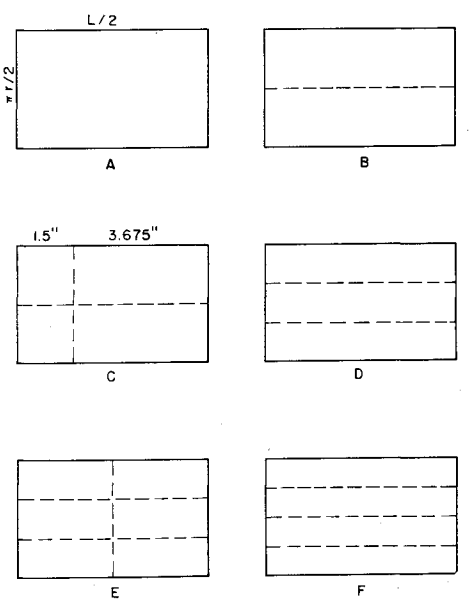


Fig. 3 Various modelings.

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Table 2 Constants and exponents for unsymmetric submatrices (continued)

I	J	(8) G IJ	(9) G IJ	(10) G IJ	(11) G IJ	(12) G IJ	(13) G IJ	(14) G IJ	L IJ	M IJ
9	1	-1/ 4	-1/ 4	-9/ 140	-9/ 140	0/ 1	-3/ 5	-3/ 5	0	0
9	2	- / 20	-1/ 20	-9/ 700	-13/ 840	0/ 1	-1/ 20	-1/ 20	1	0
9	3	-1/ 20	-1/ 20	-13/ 840	-9/ 700	9/ 70	-3/ 25	-3/ 25	0	1
9	4	-1/ 100	-1/ 100	-13/ 4200	-13/ 4200	13/420	-1/ 100	-1/ 100	1	1
9	5	1/ 4	-1/ 4	-13/ 70	9/ 140	0/ 1	-3/ 5	3/ 5	0	0
9	6	1/ 20	-1/ 20	-13/ 350	13/ 840	0/ 1	-1/ 20	1/ 20	1	0
9	7	1/ 20	1/ 20	11/ 420	9/ 700	-9/ 70	3/ 25	3/ 25	0	1
9	8	1/ 100	1/ 100	11/ 2100	13/ 4200	-13/420	1/ 100	1/ 100	1	1
9	9	1/ 4	1/ 4	13/ 70	13/ 70	0/ 1	3/ 5	-3/ 5	0	0
9	10	-1/ 20	1/ 20	13/ 350	-11/ 420	-0/ 1	-1/ 20	1/ 20	1	0
9	11	1/ 20	-1/ 20	-11/ 420	13/ 350	-13/ 35	-3/ 25	-3/ 25	0	1
9	12	-1/ 100	-1/ 100	-11/ 2100	-11/ 2100	11/210	1/ 100	1/ 100	1	1
9	13	-1/ 4	1/ 4	9/ 140	-13/ 70	0/ 1	3/ 5	3/ 5	0	0
9	14	1/ 20	1/ 20	9/ 700	11/ 420	-0/ 1	-1/ 20	-1/ 20	1	0
9	15	-1/ 20	1/ 20	13/ 840	-13/ 350	13/ 35	3/ 25	3/ 25	0	1
9	16	1/ 100	1/ 100	13/ 4200	11/ 2100	-11/210	-1/ 100	-1/ 100	1	1
10	1	1/ 20	1/ 20	9/ 700	13/ 840	-0/ 1	1/ 20	1/ 20	1	0
10	2	1/ 120	1/ 120	3/ 1400	1/ 280	-0/ 1	-1/ 60	-1/ 60	2	0
10	3	1/ 100	1/ 100	13/ 4200	13/ 4200	-13/420	1/ 100	1/ 100	1	1
10	4	1/ 600	1/ 600	13/25200	1/ 1400	-1/140	-1/ 300	-1/ 300	2	1
10	5	-1/ 20	1/ 20	13/ 350	-13/ 840	-0/ 1	1/ 20	-1/ 20	1	0
10	6	1/ 120	1/ 120	13/ 2100	-1/ 280	-0/ 1	-1/ 60	1/ 60	2	0
10	7	-1/ 100	-1/ 100	-11/ 2100	-13/ 4200	13/420	-1/ 100	-1/ 100	1	1
10	8	-1/ 600	-1/ 600	-11/12600	-1/ 1400	1/140	1/ 300	1/ 300	2	1
10	9	1/ 20	-1/ 20	-13/ 350	-11/ 420	-0/ 1	-1/ 20	11/ 20	1	0
10	10	0/ 1	0/ 1	0/ 1	1/ 210	0/ 1	1/ 15	-1/ 15	2	0
10	11	1/ 100	1/ 100	11/ 2100	-11/ 2100	11/210	1/ 100	11/ 100	1	1
10	12	0/ 1	-0/ 1	-0/ 1	1/ 1050	-1/105	-1/ 75	-1/ 75	2	1
10	13	-1/ 20	-1/ 20	-9/ 700	11/ 420	-0/ 1	-1/ 20	-11/ 20	1	0
10	14	-0/ 1	0/ 1	0/ 1	-1/ 210	0/ 1	1/ 15	1/ 15	2	0
10	15	-1/ 100	-1/ 100	-13/ 4200	11/ 2100	-11/210	-1/ 100	-11/ 100	1	1
10	16	-0/ 1	0/ 1	0/ 1	-1/ 1050	1/105	1/ 75	1/ 75	2	1
11	1	1/ 20	1/ 20	13/ 840	9/ 700	-9/ 70	3/ 25	3/ 25	0	1
11	2	1/ 100	1/ 100	13/ 4200	13/ 4200	-13/420	1/ 100	1/ 100	1	1
11	3	1/ 120	1/ 120	1/ 280	3/ 1400	-9/140	1/ 50	1/ 50	0	2
11	4	1/ 600	1/ 600	1/ 1400	13/25200	-13/840	1/ 600	1/ 600	1	2
11	5	-1/ 20	-1/ 20	11/ 420	-9/ 700	9/ 70	-3/ 25	-3/ 25	0	1
11	6	-1/ 100	-1/ 100	11/ 2100	-13/ 4200	13/420	-1/ 100	-1/ 100	1	1
11	7	0/ 1	-0/ 1	-1/ 210	0/ 1	9/140	-0/ 1	0/ 1	0	2
11	8	0/ 1	-0/ 1	-1/ 1050	0/ 1	13/840	-0/ 1	0/ 1	1	2
11	9	-1/ 20	1/ 20	-11/ 420	-13/ 350	13/ 35	3/ 25	3/ 25	0	1
11	10	1/ 100	1/ 100	-11/ 2100	11/ 2100	-11/210	-1/ 100	-1/ 100	1	1
11	11	0/ 1	0/ 1	1/ 210	0/ 1	13/ 70	0/ 1	-0/ 1	0	2
11	12	-0/ 1	0/ 1	1/ 1050	-0/ 1	-11/420	-0/ 1	0/ 1	1	2
11	13	1/ 20	-1/ 20	-13/ 840	13/ 350	-13/ 35	-3/ 25	-3/ 25	0	1
11	14	-1/ 100	-1/ 100	-13/ 4200	-11/ 2100	11/210	1/ 100	1/ 100	1	1
11	15	1/ 120	-1/ 120	-1/ 280	13/ 2100	-13/ 70	-1/ 50	-1/ 50	0	2
11	16	-1/ 600	-1/ 600	-1/ 1400	-11/12600	11/420	1/ 600	1/ 600	1	2
12	1	-1/ 100	-1/ 100	-13/ 4200	13/ 4200	13/420	-1/ 100	-1/ 100	1	1
12	2	-1/ 600	-1/ 600	-13/25200	-1/ 1400	1/140	1/ 300	1/ 300	2	1
12	3	-1/ 600	-1/ 600	-1/ 1400	-13/25200	13/840	-1/ 600	-1/ 600	1	2
12	4	-1/3600	-1/3600	-1/ 8400	-1/ 8400	1/280	1/1800	1/1800	2	2
12	5	1/ 100	1/ 100	-11/ 2100	13/ 4200	-13/420	1/ 100	1/ 100	1	1
12	6	1/ 600	1/ 600	-11/12600	1/ 1400	-1/140	-1/ 300	-1/ 300	2	1
12	7	-0/ 1	0/ 1	1/ 1050	-0/ 1	-13/840	0/ 1	-0/ 1	1	2
12	8	-0/ 1	0/ 1	1/ 6300	-0/ 1	-1/280	-0/ 1	0/ 1	2	2
12	9	-1/ 100	-1/ 100	11/ 2100	11/ 2100	-11/210	-1/ 100	-11/ 100	1	1
12	10	-0/ 1	0/ 1	-0/ 1	-1/ 1050	1/105	1/ 75	1/ 75	2	1
12	11	0/ 1	-0/ 1	-1/ 1050	-0/ 1	-11/420	-0/ 1	0/ 1	1	2
12	12	0/ 1	0/ 1	0/ 1	1/210	0/ 1	-0/ 1	0/ 1	2	2
12	13	1/ 100	1/ 100	13/ 4200	-11/ 2100	11/210	1/ 100	11/ 100	1	1
12	14	0/ 1	-0/ 1	-0/ 1	1/ 1050	-1/105	-1/ 75	1/ 75	2	1
12	15	1/ 600	1/ 600	1/ 1400	-11/12600	11/420	1/ 600	11/ 600	1	2
12	16	0/ 1	-0/ 1	-0/ 1	1/ 6300	-1/210	-1/ 450	-1/ 450	2	2

I	J	(8) G IJ	(9) G IJ	(10) G IJ	(11) G IJ	(12) G IJ	(13) G IJ	(14) G IJ	L IJ	M IJ
13	1	-1/ 4	1/ 4	-13/ 70	-9/ 140	0/ 1	3/ 5	-3/ 5	0	0
13	2	-1/ 20	1/ 20	-13/ 350	-13/ 840	0/ 1	1/ 20	-1/ 20	1	0
13	3	1/ 20	1/ 20	-11/ 420	9/ 700	-9/ 70	3/ 25	3/ 25	0	1
13	4	1/ 100	1/ 100	-11/ 2100	13/ 4200	-13/420	1/ 100	1/ 100	1	1
13	5	1/ 4	1/ 4	-9/ 140	9/ 140	0/ 1	3/ 5	3/ 5	0	0
13	6	1/ 20	1/ 20	-9/ 700	13/ 840	0/ 1	1/ 20	1/ 20	1	0
13	7	-1/ 20	-1/ 20	13/ 840	-9/ 700	9/ 70	-3/ 25	-3/ 25	0	1
13	8	-1/ 100	-1/ 100	13/ 4200	13/420	-13/ 4200	-1/ 100	-1/ 100	1	1
13	9	1/ 4	-1/ 4	9/ 140	13/ 70	0/ 1	-3/ 5	-3/ 5	0	0
13	10	-1/ 20	1/ 20	9/ 700	-11/ 420	-0/ 1	1/ 20	1/ 20	1	0
13	11	-1/ 20	1/ 20	-13/ 840	-13/ 350	13/ 35	3/ 25	3/ 25	0	1
13	12	1/ 100	1/ 100	-13/ 4200	11/ 2100	-11/210	-1/ 100	-1/ 100	1	1
13	13	-1/ 4	1/ 4	13/ 70	-13/ 70	0/ 1	-3/ 5	3/ 5	0	0
13	14	1/ 20	-1/ 20	13/ 350	11/ 420	-0/ 1	1/ 20	-1/ 20	1	0
13	15	1/ 20	-1/ 20	11/ 420	13/ 350	-13/ 35	-3/ 25	-3/ 25	0	1
13	16	-1/ 100	-1/ 100	11/ 2100	-11/ 2100	11/210	1/ 100	1/ 100	1	1
14	1	1/ 20	-1/ 20	13/ 350	13/ 840	-0/ 1	-1/ 20	1/ 20	1	0
14	2	1/ 120	-1/ 120	13/ 2100	1/ 280	-0/ 1	1/ 60	-1/ 60	2	0
14	3	-1/ 100	-1/ 100	11/ 2100	13/ 4200	-13/420	-1/ 100	-1/ 100	1	1
14	4	-1/ 600	-1/ 600	11/12600	-1/ 1400	1/140	1/ 300	1/ 300	2	1
14	5	-1/ 20	-1/ 20	9/ 700	13/ 840	-0/ 1	-1/ 20	-1/ 20	1	0
14	6	-1/ 120	-1/ 120	3/ 1400	-1/ 280	-0/ 1	1/ 60	1/ 60	2	0
14	7	1/ 100	1/ 100	-13/ 4200	13/ 4200	-13/420	1/ 100	1/ 100	1	1
14	8	1/ 600	1/ 600	-13/25200	1/ 1400	-1/140	-1/ 300	-1/ 300	2	1
14	9	1/ 20	1/ 20	-9/ 700	-11/ 420	-0/ 1	1/ 20	11/ 20	1	0
14	10	0/ 1	-0/ 1	0/ 1	1/ 210	0/ 1	-1/ 15	-1/ 15	2	0
14	11	-1/ 100	-1/ 100	13/ 4200	11/ 2100	-11/210	-1/ 100	-11/ 100	1	1
14	12	-0/ 1	-0/ 1	-0/ 1	-1/ 1050	1/105	1/ 75	1/ 75	2	1
14	13	-1/ 20	1/ 20	-13/ 350	11/ 420	-0/ 1	1/ 20	-11/ 20	1	0
14	14	-0/ 1	-0/ 1	0/ 1	-1/ 210	0/ 1	-1/ 15	1/ 15	2	0
14	15	1/ 100	1/ 100	-11/ 2100	-11/ 2100	11/210	1/ 100	11/ 100	1	1
14	16	0/ 1	-0/ 1	0/ 1	1/ 1050	-1/105	-1/ 75	1/ 75	2	1
15	1	-1/ 20	-1/ 20	-11/ 420	-9/ 700	9/ 70	-3/ 25	-3/ 25	0	1
15	2	-1/ 100	-1/ 100	-11/ 2100	-13/ 4200	13/420	-1/ 100	-1/ 100	1	1
15	3	0/ 1	-0/ 1	-1/ 210	0/ 1	-9/140	-0/ 1	0/ 1	0	2
15	4	0/ 1	-0/ 1	-1/ 1050	0/ 1	-13/840	-0/ 1	0/ 1	1	2
15	5	1/ 20	1/ 20	-13/ 840	9/ 700	-9/ 70	3/ 25	3/ 25	0	1
15	6	1/ 100	1/ 100	-13/ 4200	13/ 4200	-13/420	1/ 100	1/ 100	1	1
15	7	-1/ 120	-1/ 120	1/ 280	-3/ 1400	9/140	-1/ 50	-1/ 50	0	2
15	8	-1/ 600	-1/ 600	1/ 1400	-13/25200	13/840	-1/ 600	-1/ 600	1	2
15	9	1/ 20	-1/ 20	13/ 840	13/ 350	-13/ 35	-3/ 25	-3/ 25	0	1
15	10	-1/ 100	-1/ 100	13/ 4200	-11/ 2100	11/210	1/ 100	1/ 100	1	1
15	11	-1/ 120	1/ 120	-1/ 280	-13/ 2100	13/ 70	1/ 50	1/ 50	0	2
15	12	1/ 600	1/ 600	-1/ 1400	11/12600	-11/420	-1/ 600	-1/ 600	1	2
15	13	-1/ 20	1/ 20	11/ 420	-13/ 350	13/ 35	3/ 25	3/ 25	0	1
15	14	1/ 100	1/ 100	11/ 2100	11/ 2100	-11/210	-1/ 100	-1/ 100	1	1
15	15	0/ 1	0/ 1	1/ 210	0/ 1	-13/ 70	0/ 1	-0/ 1	0	2
15	16	-0/ 1	0/ 1	1/ 1050	-0/ 1	11/420	-0/ 1	0/ 1	1	2
16	1	1/ 100	1/ 100	11/ 2100	13/ 4200	-13/420	1/ 100	1/ 100	1	1
16	2	1/ 600	1/ 600	11/12600	1/ 1400	-1/140	-1/ 300	-1/ 300	2	1
16	3	-0/ 1	0/ 1	1/ 1050	-0/ 1	13/840	0/ 1	-0/ 1	1	2
16	4	-0/ 1	0/ 1	1/ 6300	-0/ 1	1/280	-0/ 1	0/ 1	2	2
16	5	-1/ 100	-1/ 100	13/ 4200	13/420	-13/420	1/ 100	1/ 100	1	1
16	6	-1/ 600	-1/ 600	13/25200	-1/ 1400	1/140	1/ 300	1/ 300	2	1
16	7	1/ 600	1/ 600	-1/ 1400	13/25200	-13/840	1/ 600	1/ 600	1	2
16	8	1/3600	1/3600	-1/ 8400	1/ 8400	-1/280	-1/1800	-1/1800	2	2
16	9	1/ 100	1/ 100	-13/ 4200	-11/ 2100	11/210	1/ 100	11/ 100	1	1
16	10	0/ 1	-0/ 1	0/ 1	1/ 1050	-1/105	-1/ 75	-1/ 75	2	1
16	11	-1/ 600	-1/ 600	1/ 1400	11/12600	-11/420	-1/ 600	-11/ 600	1	2
16	12	-0/ 1	0/ 1	-0/ 1	-1/ 6300	1/210	1/ 450	1/ 450	2	2
16	13	-1/ 100	-1/ 100	-11/ 2100	11/ 2100	-11/210	-1/ 100	-11/ 100	1	1
16	14	-0/ 1	0/ 1	0/ 1	-1/ 1050	1/105	1/ 75	1/ 75	2	1
16	15	0/ 1	-0/ 1	-1/ 1050	-0/ 1	11/420	-0/ 1	0/ 1	1	2
16	16	0/ 1	0/ 1	0/ 1	0/ 1	-1/210	0/ 1	-0/ 1	2	2

sidered; because of the double symmetry of the problem, only the octant indicated need be modeled. The displacement under the load is given by Timoshenko[12] as $w_p = -0.1084$ in. and is calculated using inextensional theory. The results of the various modelings shown in Fig. 3 are given in Table 3. The difference between runs 3 and 4 in Table 3 is that continuity of the derivatives of the membrane displacements u and v was imposed between adjacent elements in run 3 but not in run 4. It is important to realize that, in order to maintain geometric admissibility, the derivatives of the transverse displacement must be forced to be continuous; however, only the membrane displacements and not their derivatives need be continuous between elements.

In assessing monotonic convergence, only the results of those modelings that are successive refinements of each other should be compared. Therefore, comparing $\min\Pi_p$ of runs 1, 2, 3, 4, runs 1, 2, 7, and runs 1, 5, 6, it is noted that monotonic convergence is exhibited in each case. In any finite-element solution, the accuracy of the results depends to a certain extent upon the modeling selected. This is seen in the present problem by observing that a lower $\min\Pi_p$ was obtained in run 7 with 72 degrees of freedom than in run 6 with 90 degrees of freedom, and similarly with runs 4 and 5.

Table 3 Numerical results

Run	Element modeling	Degrees of freedom	Min. Π_p, in.-lb	w_p, in.
1	*A*	18	−0.0327	−0.0025
2	*B*	36	−1.0022	−0.0802
3	*C*	60	−1.0104	−0.0808
4	*C*	97	−1.0636	−0.0851
5	*D*	54	−1.2827	−0.1026
6	*E*	90	−1.2944	−0.1036
7	*F*	72	−1.3588	−0.1087

An inextensional solution yields results that are too stiff, which explains why the displacement of run 7 is larger than that given by Timoshenko. As the shell is made thinner, the inextensional theory becomes more nearly exact. This is indicated by considering the same cylinder as the foregoing but taking only half the thickness; in this case, the inextensional theory gives $w_p = -0.8674$ in., whereas modeling F yields $w_p = -0.8322$ in.

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APRIL 1967 AIAA JOURNAL VOL. 5, NO. 4

Inextensional Buckling Configurations of Conical Shells

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Relations describing the geometrical forms taken by a thin conical shell when it shortens under axial compression by pure folding are presented. These forms are polyhedral surfaces characterized by two distinct sets of triangular planes, one rotated toward [called the (+) surface] and the other away from [called the (−) surface] the generatrix of the undeformed cone. The number of circumferential waves in each row remains constant along the length of the shell, with the size varying according to the cone diameter. The (+) and (−) surfaces have distinct aspect ratios, the (−) surface being always the more slender in the circumferential direction. The relations are valid for all degrees of axial shortening including that corresponding to total collapse. Collapse patterns obtained from experiments are shown to agree qualitatively with those constructed from the theory.

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Nomenclature

α	=	angle of rotation of (+) planes
β	=	angle of rotation of (−) planes
γ	=	semivertex angle of cone
δ	=	total axial shortening corresponding to one row of half-waves
δ_0	=	outward radial displacement of a vertex from the reference circle
δ_i	=	maximum inward radial displacement of the cross-sectional edge of the buckled form from the reference circle
ϵ	=	unit axial shortening or strain
$\theta/2$	=	$(\pi/n)\sin\gamma$
D	=	diameter of the reference circle of the cone
g, h	=	lengths defined in Fig. 2
$K = l_y/l_x$	=	aspect ratio of buckle waves
l_x	=	axial half-wavelength
l_y	=	circumferential half-wavelength
n	=	number of full waves in one circumference

Superscripts

$(^+)$	=	(+) plane
$(^-)$	=	(−) plane
$(\ ')$	=	the cross section adjacent to the reference cross section

Introduction

THE asymmetrical buckled form assumed by a thin cylindrical shell under axial compression was shown by Yoshimura[1] to be a polyhedral surface consisting of an array of equal triangular planes. In this configuration the shell is shortened axially from the initial form almost totally by folding, and hence the state of deformation is essentially inextensional. The geometry in the perfectly inextensional state is described[2] by the relation

$$\epsilon = 1 - [1 - K^2 \tan^2(\pi/2n)]^{1/2} \tag{1}$$

where ϵ is the unit axial shortening, K, the aspect ratio of the buckle waves (K = circumferential half-wavelength/axial half-wavelength) and n the number of circumferential waves. The present writer showed that thin cylindrical shells under axial impact at low velocity collapse in configurations described approximately by

$$K = \frac{1}{\tan(\pi/2n)} \tag{2}$$

which is obtained from Eq. (1) by setting $\epsilon = 1$. Similar folding configurations exist for conical shells and relations describing them are derived in this paper.

Analysis

For perfect inextensional buckling under axial compression, it is required that the axial shortening of the shell be produced by a bending of the surface about straight lines in such a way that extensional strain in the middle surface is zero everywhere. We consider first the over-all inextensional buckling configuration for the conical shell and show, in Figs. 1a and 1b, the horizontal and vertical views, respectively, of the initial and buckled configurations. The conical frustum has a semivertex angle γ and upper and lower bases A and C, respectively, which are assumed to be free of lateral restraint. The buckled configuration is represented by n circumferential waves and one axial wave, although any number of axial waves may be considered as well. As a result of buckling, circular sections such as A, B, and C become regular polygons of unchanged perimeter. Several vertices

Presented as Preprint 66–125 at the AIAA 3rd Aerospace Sciences Meeting, New York, January 24–26, 1966; submitted February 3, 1966. This work was partially sponsored by the Air Force Flight Dynamics Laboratory, Research and Technology Division, Air Force Systems Command under Contract AF 33(657)-10220. [6.02, 6.04]

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